

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

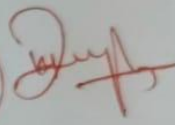
COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

95
100



QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I N. Aima Harshitha ⁽¹⁹²³⁷¹⁰⁶³⁾ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N. Aima Harshitha

Question 1

Responsive Web Design (RWD)

Responsive Web Design (RWD) is a web development approach that enables a website to automatically adjust its layout and content according to different screen sizes such as mobile phones, tablets, and desktops. The main goal is to provide an optimal viewing experience without creating separate websites for different devices.

Importance

- Improves user experience on all devices.
- Reduces maintenance by using one website for all screen sizes.
- Increases accessibility and readability.
- Improves SEO because search engines prefer mobile-friendly websites.

CSS Techniques for Responsiveness

1. **Fluid Layouts** – Use percentage (%) instead of fixed pixel widths.
2. **Flexible Images** – Set images with max-width: 100%.
3. **Media Queries** – Apply different CSS rules based on screen size.
4. **Viewport Meta Tag** – Makes the webpage adapt to the device width.

Using Flexbox/Grid

CSS Flexbox and CSS Grid help create adaptive layouts that automatically arrange elements based on available space.

Example (Flexbox):

```
<div class="container">
```

```
<div>Item 1</div>
```

```
<div>Item 2</div>
```

```
<div>Item 3</div>
```

```
</div>
```

```
.container{
```

```
display:flex;
```

```
    flex-wrap:wrap;
}

.container div{
    flex:1 1 200px;
    margin:10px;
    background:lightblue;
    padding:20px;
}
```

The layout automatically adjusts the number of items per row depending on the screen width.

Improvements

- Use responsive images.
- Increase button size for touch devices.
- Optimize fonts for readability.
- Test the website on different devices and browsers.

Question 2

Client-side Validation

Client-side validation checks user input in the browser before the form is submitted to the server. It provides immediate feedback and reduces unnecessary server requests.

Purpose

- Prevent invalid data submission.
- Improve user experience.
- Reduce server load.
- Save user time.

JavaScript Event Handling

JavaScript events such as submit, input, and blur trigger validation functions.

Role of Regular Expressions (Regex)

Regular expressions are patterns used to verify whether input follows a required format.

Example:

```
<form onsubmit="return validateForm()">
<input type="email" id="email" placeholder="Enter Email">
<input type="submit">
</form>

function validateForm(){
    let email=document.getElementById("email").value;
    let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if(!pattern.test(email)){
        alert("Invalid Email");
        return false;
    }
    return true;
}
```

A phone number can also be validated using a regex such as:

```
/^[0-9]{10}$/
```

Improvements

- Display error messages beside the input field.
- Use HTML5 validation attributes like required and pattern.
- Always perform server-side validation for security.
- Sanitize user inputs to prevent attacks.

Question 3

CSS Box Model

The CSS Box Model describes how elements occupy space on a webpage.

Components

1. **Content** – The actual text or image.
2. **Padding** – Space around the content.
3. **Border** – Surrounds the padding.
4. **Margin** – Space outside the border.

CSS Positioning Techniques

- **Static** – Default positioning.
- **Relative** – Positioned relative to its normal position.
- **Absolute** – Positioned relative to the nearest positioned ancestor.
- **Fixed** – Stays fixed during scrolling.
- **Sticky** – Switches between relative and fixed.

Causes of Layout Breaking

- Fixed-width elements.
- Large images without scaling.
- Incorrect positioning.
- Overflowing content.
- Missing media queries.

Using Media Queries

```
@media(max-width:768px){  
.container{  
    flex-direction:column;  
}  
}
```

This changes the layout from horizontal to vertical on smaller screens.

Best Practices

- Use Flexbox or Grid.
- Avoid fixed widths.
- Use box-sizing: border-box.
- Optimize images.
- Test layouts on multiple screen sizes.

Question 4

Document Object Model (DOM)

The Document Object Model (DOM) is a tree-like representation of an HTML document. It allows JavaScript to access and modify webpage elements dynamically.

Role in Web Development

The DOM enables developers to:

- Change text and images.
- Modify CSS styles.
- Add or remove elements.
- Respond to user actions.

DOM Manipulation Example

```
<p id="demo">Welcome</p>
```

```
<button onclick="changeText()">Click</button>
```

```
function changeText(){  
    document.getElementById("demo").innerHTML="Welcome to JavaScript!";  
}
```

When the button is clicked, the paragraph text changes without reloading the webpage.

Improvements

- Use event listeners instead of inline events.
- Update only required elements to improve performance.
- Minimize unnecessary DOM manipulation.
- Use asynchronous requests (AJAX/Fetch API) for dynamic data loading.
- Provide loading indicators for better user experience.